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Studierichting: Informatica

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$$\int_2^9 \frac{3x}{\sqrt[3]{x-1}} dx$$

Stel $t = x - 1 \Leftrightarrow x = t + 1 \Rightarrow dx = dt$.

$$\begin{aligned} \Rightarrow \int_2^9 \frac{3t+3}{t^{\frac{1}{3}}} &= 3 \int_2^9 t^{\frac{2}{3}} + 3 \int_2^9 t^{\frac{-1}{3}} \\ &= 3 \left[\frac{3t^{\frac{5}{3}}}{5} + \frac{3t^{\frac{2}{3}}}{2} \right]_2^9 = \left[\frac{6\sqrt[3]{(x-1)^5} + 15\sqrt[3]{(x-1)^2}}{10} \right]_2^9 \\ &= 3 \left(\frac{6 \cdot 32 + 15 \cdot 4}{10} - \frac{6 + 15}{10} \right) = \frac{693}{10} \end{aligned}$$

References